SHAHZAIB KHAN

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SUMMARY

Backend Engineer with a solid computer science foundation from FAST-NUCES, currently pursuing a bachelor's degree at DITE (UoG). Specialized in building modular RESTful APIs, relational data models (MySQL, PostgreSQL), and backend business logic using Python (Flask/FastAPI), PHP, and Node.js. Maintainer of an open-source **custom STL containers C++ library**, leveraging AI-assisted tools for rapid interface prototyping while focusing on delivering deterministic data pipelines, robust schema constraints, secure backend services, and reliable containerized deployments.

EDUCATION

Denning Institute Of Technology & Entrepreneurship (DITE) — **Sep 2026 – Present**
BSc (Hons) Computing — University of Greenwich (UK) | 1st Year

National University of Computer and Emerging Sciences (FAST) — **Aug 2019 – Dec 2025**
BS Computer Science | 104 credits completed

Govt. Degree Science & Commerce College (Gulshan) — **Apr 2017 – Apr 2019**
Inter (HSSC) — FSc Pre-Engineering | B grade (66.55%)

Little Folks Paradise School (LFPS) — **Mar 2015 – Mar 2017**
Matric (SSC) — Science Group | A-1 grade (80.58%)

EXPERIENCE

Software Engineer (Remote) — DigiFlex Solutions — **Feb 2026 – Sep 2026**

- Architected and maintained modular backend services using Python (Flask) and PHP, building session-based authentication and structured RESTful endpoints.
- Designed and optimized relational schemas in MySQL, defining foreign key constraints, indexes, and transactional state logic for multi-step workflows.
- Integrated third-party webhooks and developed automation routines using n8n and Python scripts to process asynchronous data feeds.
- Configured backend development environments and application runtimes on Linux, maintaining local container setups with Docker.

Software Developer (Intern) — Calyx Technologies — **Aug 2025 – Nov 2025**

- Contributed to backend module design, API integrations, and database schema validation for web-based inventory and management services.
- Maintained Linux staging deployment scripts, web server routing configurations, and environment secret isolation routines.
- Standardized dependency management and team version control practices, assisting in building automated CI/CD staging pipelines.

HIGHLIGHTED PROJECTS

Booking Management System (BMS) — Workflow Engine & Invoicing (PHP)

- Engineered an enterprise booking data architecture decoupling core calculation logic from presentation formatting.
- Automated deterministic invoice artifact generation, replacing fragmented manual entry with transactional data integrity.
- Implemented role-based CRUD workflows backed by MySQL relational tables with strict referential constraints.

Automated Deployment Daemon & API Orchestration System (Flask)

- Developed a script-driven CI/CD deployment architecture capable of executing single-command software updates securely on live Linux VPS servers.
- Secured application boundaries with systemd daemon processes, Nginx reverse proxy routing, and automated SSL provisioning.
- Architected a custom interaction layer that decoupled UI rendering from third-party API response structures (Google Custom Search API).

QR Attendance — Rapid Prototype (Lovable, TypeScript)

- Scaffolded an end-to-end attendance verification app using Lovable (AI generator), wiring the generated UI to a Supabase (PostgreSQL) backend.
- Modified generated backend functions and database schema to store classroom GPS coordinates and compare student submission locations.
- Customized database constraints in PostgreSQL to prevent duplicate student check-ins and configured roster exports to CSV/Excel.

Abbas Brothers — Turnkey Industrial Website & Serverless CMS (Vanilla JS, Vite build)

- Designed and delivered a production-ready industrial website with a decoupled source/distribution architecture, native Web Components, Vite builds, GitHub Actions CI/CD, and Cloudflare Pages/Workers.
- Built a serverless content pipeline using Google Drive, Google Sheets, and Apps Script to publish equipment and spare-parts catalogs as live JSON, with automated deployment, secure contact-form routing, and edge security controls.

Web Tools Dashboard — Developer Utilities (React)

- Built a serverless, local-first web tools dashboard (using React + Tailwind CSS) for managing and publishing custom utilities, with tag search, dark mode, and usage-based sorting.
- Implemented client-side GitHub API integration and automated GitHub Pages deployment using Vite build tooling.

TECHNICAL SKILLS

- **Backend & APIs:** Python (Flask, FastAPI), PHP, REST APIs, Modular Architecture, MVC, Session Authentication
- **Languages:** Python, C++, TypeScript, JavaScript, PHP, SQL, Bash/Shell, C
- **Web Development:** React, Next.js, Node.js, Vite, Tailwind CSS, Web Components, Cloudflare Pages/Workers, Google Apps Script, GitHub API/Pages
- **Databases & Data:** MySQL, PostgreSQL (Supabase/RLS), MongoDB, Schema Design, Indexing, Transactions, In-Memory Data Structures
- **Infrastructure & DevOps:** Ubuntu, Docker, Ansible, WSL2, systemd, Nginx, SSL Automation, GitHub Actions, Git/GitLab
- **Cloud & Automation:** Cloudflare R2, DNS Management, n8n Orchestration
- **AI-Assisted Development:** Cursor, Lovable, Agentic Scaffolding, Prompt Engineering

CERTIFICATIONS

- Crash Course on Python - Google / Coursera
- Using Python to Interact with the Operating System - Google / Coursera
- Introduction to Git and GitHub - Google / Coursera
- The Unix Workbench - Johns Hopkins University / Coursera